



PubTrail

Your Night, Seamlessly Documented in Real Time

Filip Todosov

Bogdan Stefanescu

Pavlos Spanoudakis

THE PROBLEM

Forgotten Memories

Have you ever struggled to recall exactly where you went or who you were with during a long night out? Pub crawls are a classic example of these hazy, multi-stop adventures.

Scattered photos, manual notes, and coordinating across multiple apps ruins the fun. We needed a centralized, user-friendly solution.



Introducing PubTrail

Real-time Pub Crawl Tracking & Collaboration



PubTrail

CORE LIVE FEATURES



Live Timeline

Add new stops instantly as you move. Watch your path being drawn live on the Mapbox-integrated screen.



Quick Join

Collaborate with everyone! Others join your crawl simply by scanning a unique QR code.



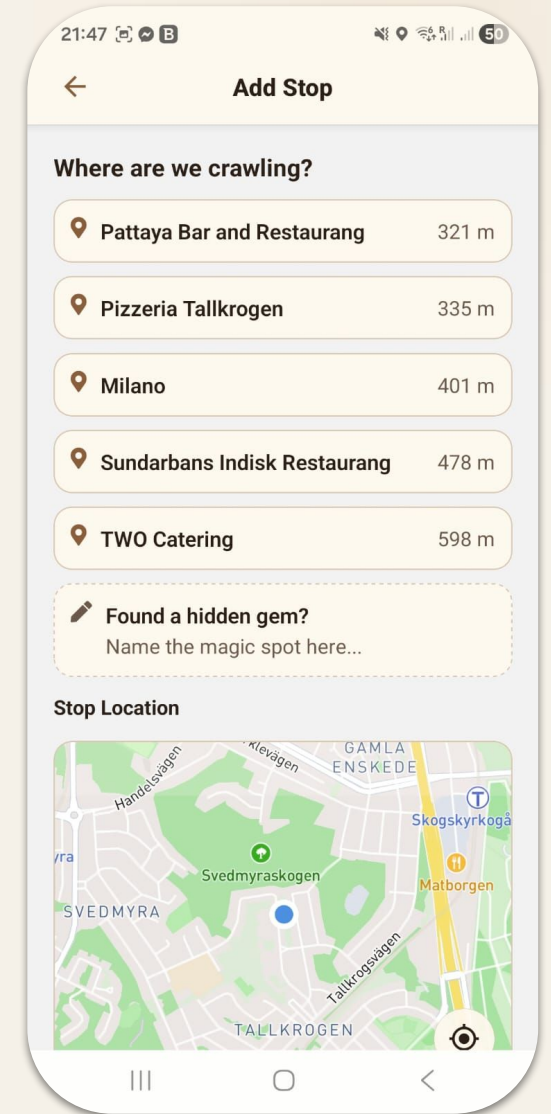
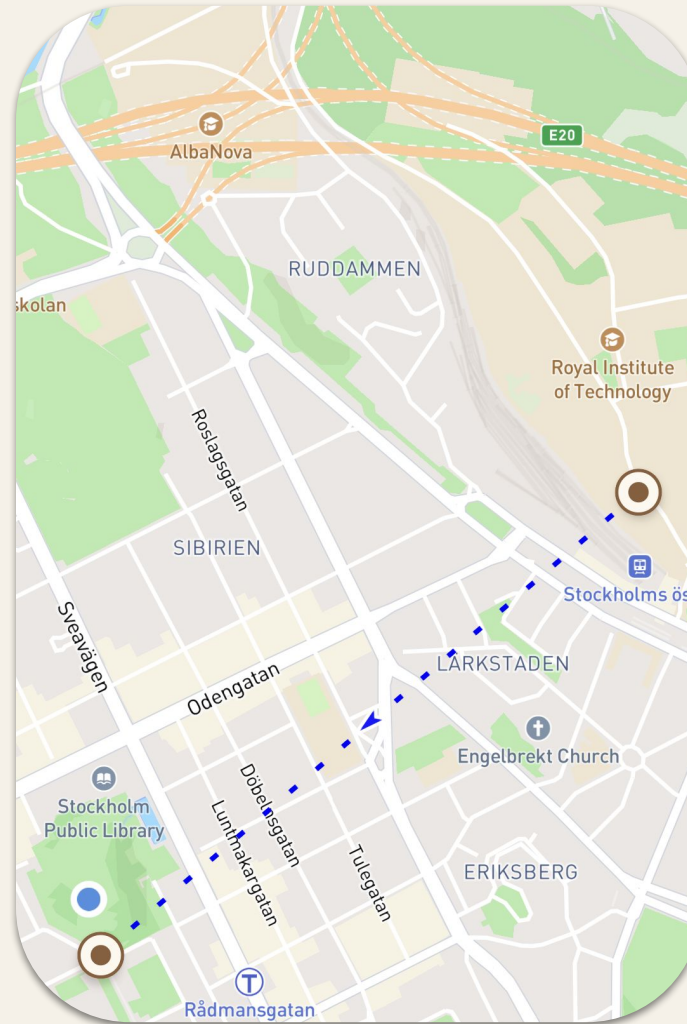
Shared Media

Upload photos to a common gallery. See live updates of everyone's best moments in one place.

REAL-TIME MAP

Our map-oriented home screen ensures you never lose track of your progress. Using Mapbox API, we provide live path drawing and nearby place suggestions for seamless stops.

Bragging rights included: Your full path is saved in your history forever.



SOCIAL DISCOVERY

Showcase Crawl 2026

Ask a friend to scan this QR code to join your crawl.



Find the Party

Don't have a group? Check for live crawls nearby in real time. If you find one, simply scan a participant's QR code and join the adventure.

History is preserved even if you leave or switch crawls, ensuring your student life legacy is always accessible.

THE SOBRIETY CHALLENGE



Bonus Feature: Balance Test

Test your sobriety with a built-in game. Utilizing your device's **gyroscope sensor**, you must maintain your balance to score high.

It's harder than it looks, especially after a few stops!
A fun way to check in with yourself before the next pub.

TECHNICAL FOUNDATION



Frontend

React Native with **TypeScript**, Expo (Router, Camera, Sensors), and **Redux** for state management.



APIs

Firebase Authentication for secure access & **Mapbox API** for sophisticated mapping and geolocation.



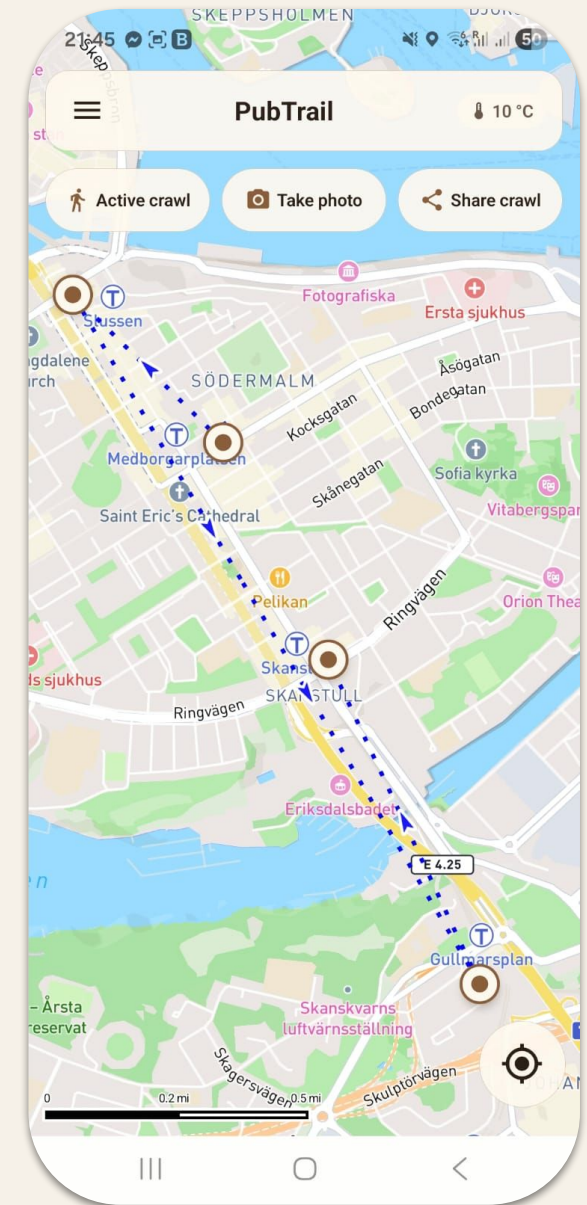
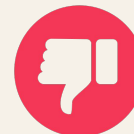
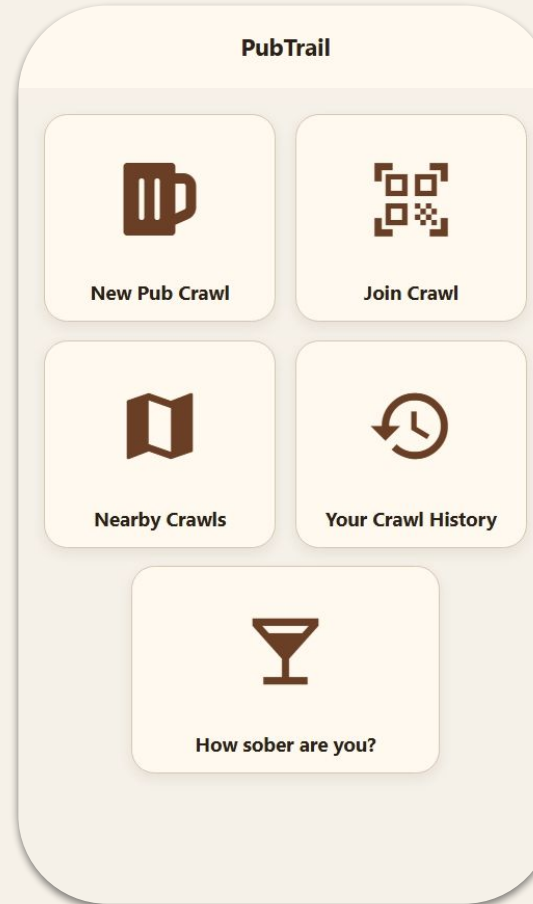
Persistence

Firestore NoSQL Database optimized with subcollections for high scalability and performance. **Cloudinary** is used to store media uploaded by users.

UI/UX STRATEGY

Intuitive, Map – Oriented UI

- Map-Oriented home screen layout instead of tile-based. If the user is in a crawl, they can easily see their crawl drawn on the map.
- Quick action buttons allow quick interactions with our crawl (taking photos, sharing via QR).



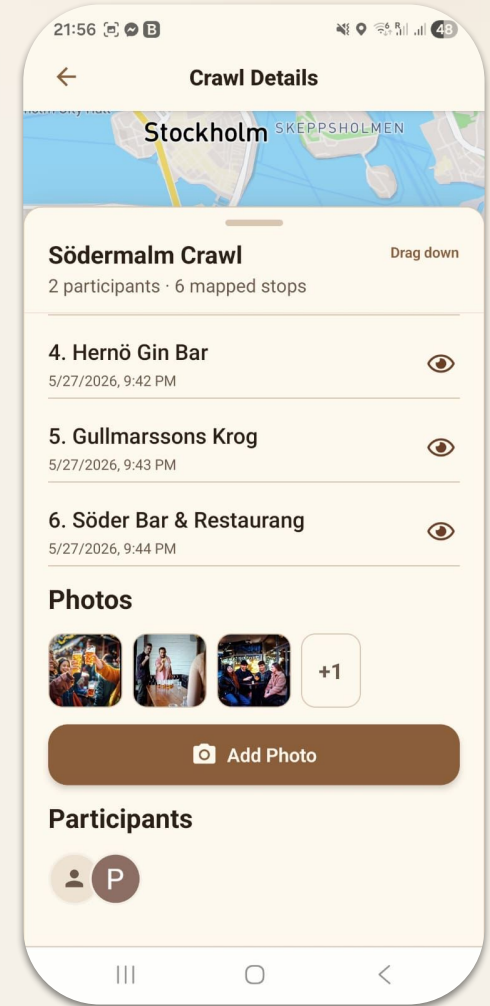
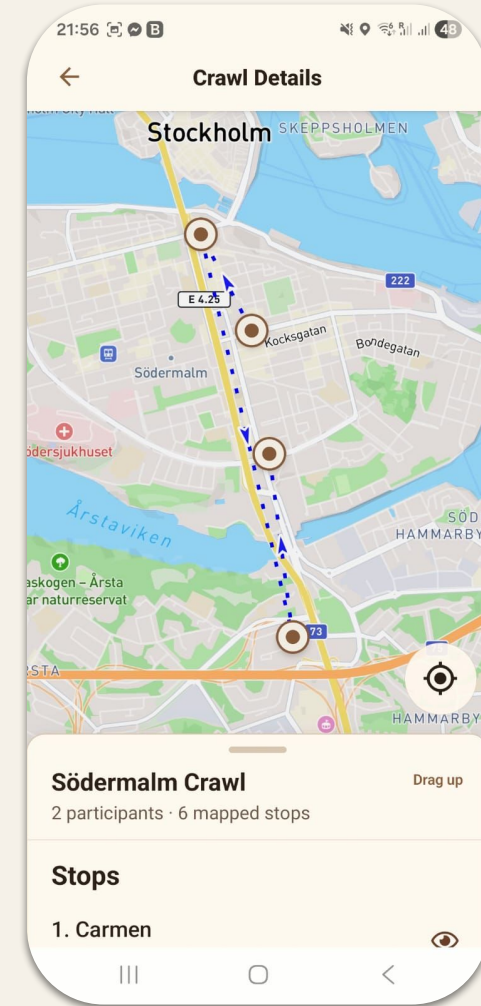
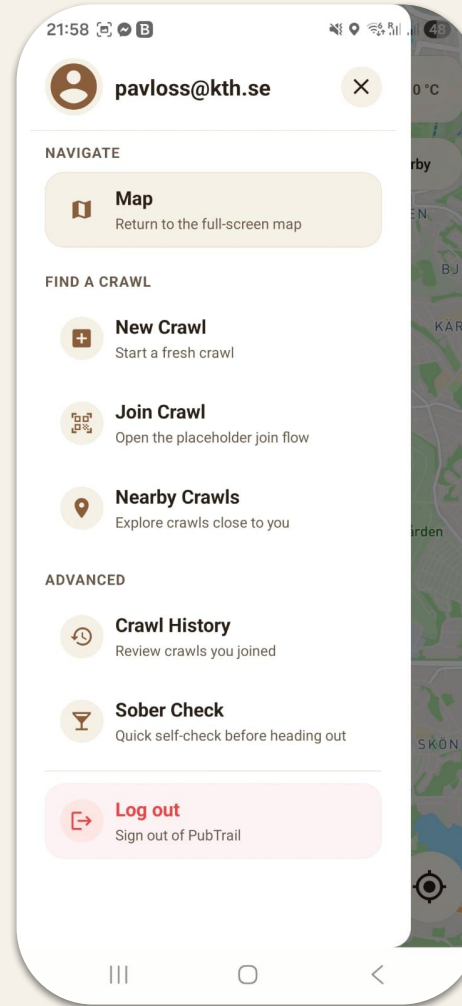
UI/UX STRATEGY

Drawers & Gesture-based menus

- » Drawer menus & gesture-based panels provide an intuitive way to navigate throughout the available options, while saving screen space for the map.

Minimal UI

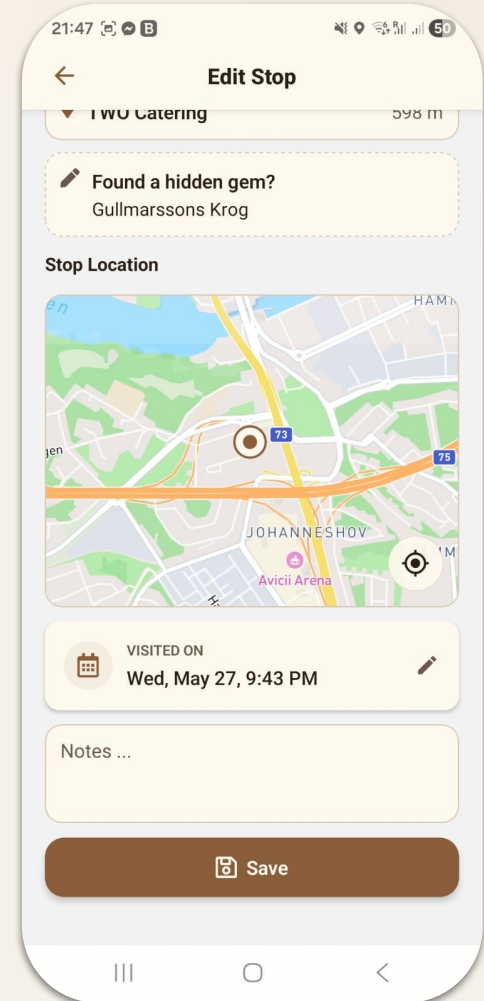
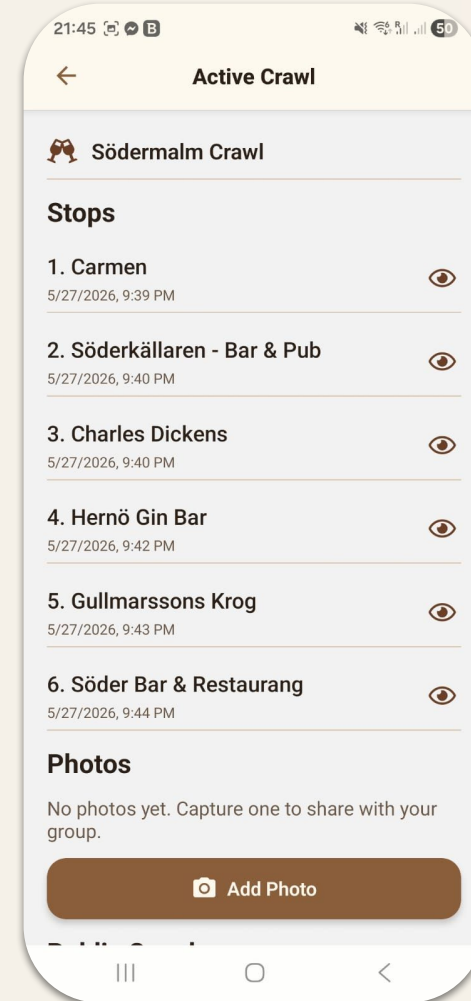
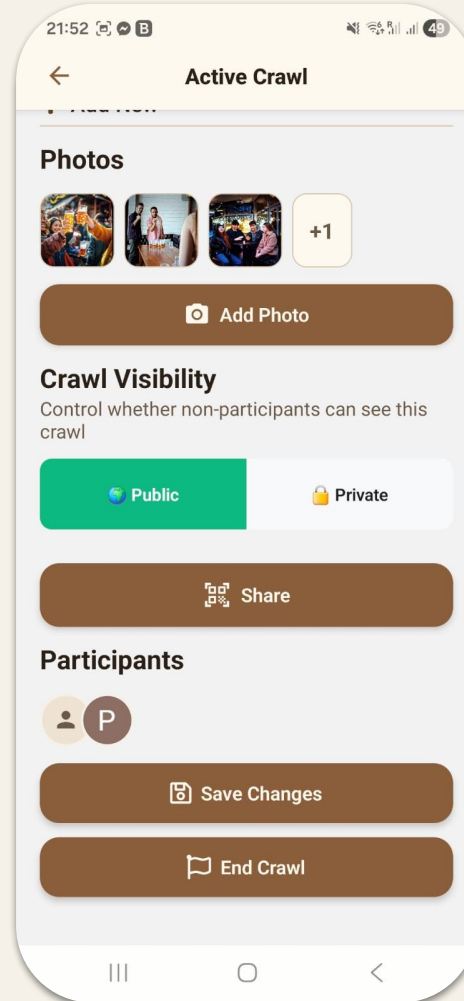
- » We avoided displaying too much information in the same screen, and kept the UI as minimal as possible without omitting crucial information.



UI/UX STRATEGY

Predictable , consistent color scheme

- ▶ We chose and applied a consistent color scheme on our UI components.
- ▶ We tried to use predictable colors on high-importance UI elements, like buttons and pressable icons.



PERSISTENCE SCHEMA

Collection	Subcollections	Key Features
Users	—	activeCrawl reference, profile info
Crawls	Stops, Media	Live location coordinates, participant list, basic crawl details
—	Stops	Location name, coordinates, timestamp, notes
—	Media	Upload URL, Uploader user reference

By using subcollections, we bypass the 1MB document limit and ensure faster loading times when fetching main collection documents (subcollections do not have to be always loaded).

Questions?

Thank you for exploring PubTrail